

Thermomechanical Constitutive Modeling in Shape Memory Polymer of Polyurethane Series

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ABSTRACT: In order to describe the thermomechanical properties in shape memory polymer of polyurethane series, a thermomechanical constitutive model was developed by modifying a standard linear viscoelastic model. The model involved a slip element due to internal friction and took account of thermal expansion. In order to describe the variation in mechanical properties due to the glass transition, coefficients in the model were expressed by a single exponential function of temperature. Several kinds of thermomechanical tests were carried out. The proposed theory expressed well the thermomechanical properties of the material, such as shape fixity, shape recovery and recovery stress. The proposed model is useful for design of shape-memory polymer elements, in which the amount of recovery deformation, the tightening force and the working start and completion temperatures are specified.

INTRODUCTION

THE development of applications of shape memory polymers (SMPs) is currently of great interest (Hayashi, 1993; Liang, Rogers and Malafeew, 1991; Hayashi, Ishikawa and Jiordano, 1991). Because SMPs are light, high in shape recovery, easy to manipulate and economical compared with shape memory alloy, their development is being actively promoted. The characteristics such as shape recovery and shape fixity exist due to the difference in mechanical properties of materials above and below the glass transition temperature (Tobushi et al., 1996b; Takahashi, Hayashi and Hayashi, 1996). The glass transition temperature T_g in the SMP of polyurethane series can be set anywhere in the range of room temperature ± 50 K, making a wide range of applications possible.

In order to design SMP elements, the constitutive equation which expresses thermomechanical properties of SMPs is necessary. A standard linear viscoelastic (SLV) model expresses well the overall mechanical properties of polymers (Lockett, 1972). The SLV model, however, cannot describe the facts that the mechanical properties vary greatly in the glass transition region and that creep recovery is different above and below T_g . In order to express the difference of creep recovery above and below T_g , a slip mechanism due to internal friction is considered. A four element model which

involves the slip mechanism is proposed. In order to describe the variation in mechanical properties due to the glass transition, coefficients in the model are expressed by a single exponential function of temperature. Furthermore the model takes account of thermal expansion.

The proposed model is evaluated by carrying out creep test, stress relaxation test and thermomechanical test. The proposed theory expresses well the thermomechanical properties of the material, such as shape fixity, shape recovery and recovery stress.

MODELING OF THERMOMECHANICAL CONSTITUTIVE EQUATION

SLV Model

In order to describe the deformation properties of polymers, viscoelastic models are used. The SLV model is a three-element model. The SLV model combines in a parallel fashion a spring and a Maxwell model in which a spring and a dashpot are arranged in a series order (Lockett, 1972). The basic deformation behavior which depends on time can be well described by the SLV model. In this model the stress-strain relationship is expressed as follows:

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\mu} - \frac{\varepsilon}{\lambda} \quad (1)$$

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where σ and ϵ denote stress and strain, respectively. E , μ and λ represent elastic modulus, viscosity and retardation time, respectively. The dot denotes time derivative.

The relationship between creep strain and time for SMP (Tobushi et al., 1996a) is schematically shown in Figure 1. As shown in Figure 1, the creep strain ϵ_c is recovered at temperatures T above T_g . On the contrary, ϵ_c is not recovered at $T < T_g$. The creep irrecovery strain ϵ_s increases in proportion to ϵ_c . As found in Equation (1), creep strain is perfectly recovered under $\sigma = 0$. Therefore Equation (1) does not express creep irrecovery strain at $T < T_g$. The creep irrecovery strain is expressed by a Burgers model, a four-element model, where a Maxwell and a Kelvin model are connected in series (Findley, Lai, and Onaran, 1976). Although the creep irrecovery strain of SMP at $T > T_g$ is expressed by the Burgers model with a linear dashpot, the creep irrecovery strain of SMP at $T < T_g$ cannot be expressed by the Burgers model.

Model Having Slip Mechanism Due to Internal Friction

The relationships between creep strain ϵ_c and creep irrecovery strain ϵ_s obtained by creep tests at temperatures $T_h = T_g + 20$ K and $T_l = T_g - 20$ K are shown in Figure 2. Although ϵ_s is small at T_h , ϵ_s is large at T_l . If ϵ_c is larger than a certain value ϵ_b , ϵ_s increases in proportion to ϵ_c . This relationship is expressed by the following equation

$$\epsilon_s = C(\epsilon_c - \epsilon_l) \quad (2)$$

where ϵ_l and C depend on temperature.

These creep deformation behaviors can be explained by a four-element model shown in Figure 3. In Figure 3, E_1 and E_2 denote linear springs and μ a dashpot. As seen in Figure 2, if ϵ_c is larger than the critical value ϵ_b , creep strain ($\epsilon_c - \epsilon_l$) is not recovered by a certain ratio C . This irrecoverable phenomenon can be explained by a slip element due to internal friction shown in Figure 3. The irrecoverable strain appears due to slip. The slip may appear due to orientation of molecular chain and decoupling of cross-link. The slip does not appear by internal friction till the certain critical value ϵ_l . When slip appears by the force larger than the internal friction, a certain ratio of creep strain remains as irrecoverable strain ϵ_s . Because micro-Brownian motion of molecular chain is activated at high temperatures T_h above T_g , large creep strain appears under low stress and the strain is recovered. Therefore internal friction is small at T_h , resulting in large ϵ_l and small C . On the contrary, because micro-Brownian motion of molecular chain is frozen at low temperatures T_l below T_g , creep strain appears under high stress and the strain is not recovered. Therefore internal friction is large at T_l , resulting in small ϵ_l and large C .

The stress-strain relationship in the four-element model shown in Figure 3 is expressed as follows

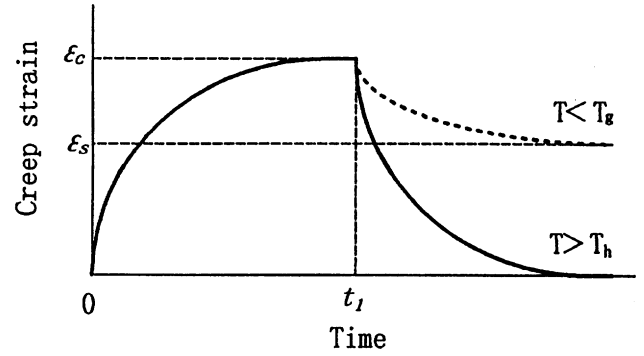


Figure 1. Schematic diagram of creep and creep recovery.

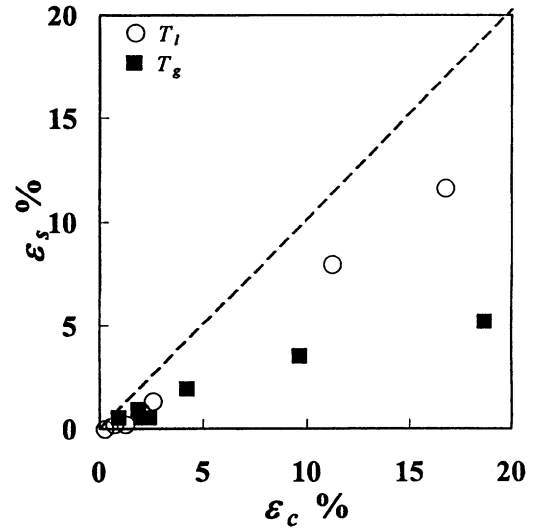


Figure 2. Relationship between creep irrecovery strain and creep strain.

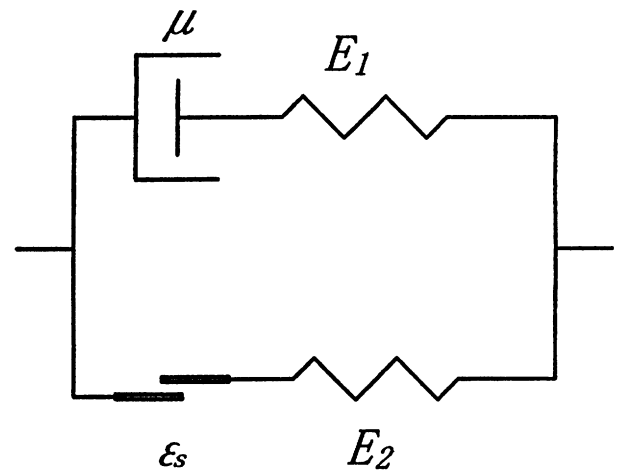


Figure 3. Four-element model.

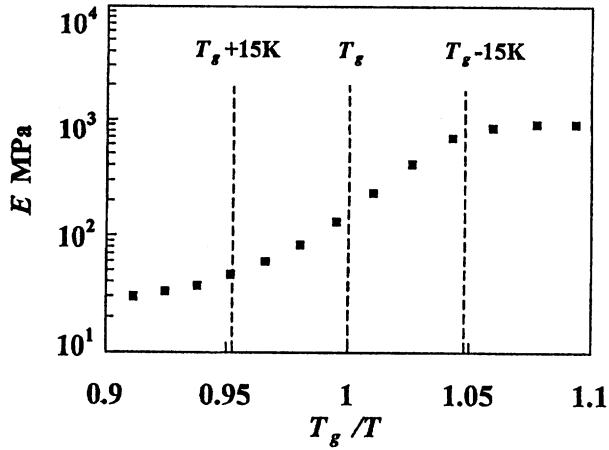
$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\mu} - \frac{\varepsilon - \varepsilon_s}{\lambda} \quad (3)$$

Constitutive Equation Considering Thermal Expansion

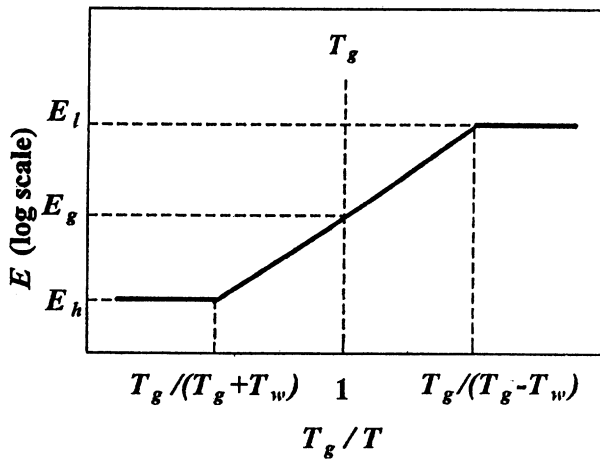
If SMP is cooled under constraint of strain, stress increases (Tobushi et al., 1996a). The material expands due to increase in temperature. These behaviors appear to be due to thermal expansion. Let us consider the thermal expansion which is assumed to be independent on the mechanical behaviour shown by Equation (3). Therefore the stress-strain-temperature relationship is

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\mu} - \frac{\varepsilon - \varepsilon_s}{\lambda} + \alpha \dot{T} \quad (4)$$

where T and α denote temperature and coefficient of thermal expansion, respectively.



(a) Experimental result



(b) Approximate curve

Figure 4. Relationship between elastic modulus and temperature: (a) experimental result and (b) approximate curve.

Dependence of Coefficients on Temperature

In SMPs, the mechanical properties vary significantly in the glass transition region. Let us consider the method to express the dependence of coefficients involved in Equation (4) on temperature.

The relationship between elastic modulus E and temperature T obtained by the dynamic mechanical test is shown in Figure 4(a). In Figure 4(a), E is expressed in logarithmic scale and T in a reciprocal normalized by T_g . As seen in Figure 4(a), E changes at temperatures in the vicinity of T_g but is almost constant above and below the glass transition region. The relationship between E and T is expressed by the relation shown in Figure 4(b). The relationship between $\log E$ and T_g/T is represented by a straight line in the glass transition region $T_g - T_w \leq T \leq T_g + T_w$. The values of E above and below the glass transition region are represented by E_h and E_l , respectively. The relationship in the glass transition region is expressed as follows

$$\log E - \log E_g = a_E \left(\frac{T_g}{T} - 1 \right) \quad (5)$$

Therefore E is

$$E = E_g \exp \left\{ a_E \left(\frac{T_g}{T} - 1 \right) \right\} \quad (6)$$

In Equation (5), E_g is the value of E at $T = T_g$ and a_E denotes a slope of the straight line. The Equation (6) by an exponential function is similar to the equation for viscosity which was theoretically derived based on molecular motion by Eyring and empirically derived by Arrhenius.

In SMP, μ , λ , C and ε_l vary depending on temperature. The variation in these values is based on the glass transition in the similar manner as E . The relationship between $\log C$ and T_g/T obtained by the creep test is expressed by a straight line. Therefore it is assumed that each coefficient is expressed by an exponential function of T similar to Equation (6). Because ε_l is small at low temperature and large at high temperature, the dependence on temperature is reverse to other coefficients. The value of α is assumed to be constant. Therefore, the coefficients are expressed by the following equations:

$$\mu = \mu_g \exp \left\{ a_\mu \left(\frac{T_g}{T} - 1 \right) \right\} \quad (7)$$

$$\lambda = \lambda_g \exp \left\{ a_\lambda \left(\frac{T_g}{T} - 1 \right) \right\} \quad (8)$$

$$C = C_g \exp \left\{ a_C \left(\frac{T_g}{T} - 1 \right) \right\} \quad (9)$$

$$\varepsilon_l = \varepsilon_g \exp \left\{ -a_\varepsilon \left(\frac{T_g}{T} - 1 \right) \right\} \quad (10)$$

where μ_g , λ_g , C_g and ε_g denote values at $T = T_g$ and a_μ , a_λ , a_C and a_ε are slopes of each straight line on semi-logarithmic diagrams.

EXPERIMENTAL METHOD

The material was an SMP thin film of polyurethane of the polyester polyol series. The casting film was made from solution (Diary MS5510; produced by Mitsubishi Heavy Industries, Ltd.) under drying conditions. The glass transition temperature T_g was about 328 K. The thickness of the specimen was about 70 μm . The width was 5 mm, gauge length 25 mm and total length 75 mm.

The tests were: (1) dynamic mechanical test, (2) thermal expansion test, (3) creep test, (4) stress relaxation test, (5) thermomechanical test and (6) recovery stress test. Every test was carried out in the temperature region at T_g and above and below T_g . In the tests, tensile properties below yield stress were examined. In the tests (3) and (4), each holding time under load and no-load was 120 min. In the tests (5) and (6), loading-unloading and heating-cooling processes were carried out. In the test (5), the shape fixity due to cooling and the shape recovery due to heating were examined. In the test (6), recovery stress due to heating under constant strain was examined. In the tests (3)–(6), strain rate was 5–50%/min, and heating-cooling rate 4 K/min.

RESULTS AND DISCUSSION

Dependence of Coefficients on Temperature

Based on the tests (1)–(6), each coefficient was determined to express the thermomechanical properties. From the results of the tests (1)–(6), it is found that the behavior of strain and stress varies significantly in the temperature region $T_g - 15 \text{ K} \leq T \leq T_g + 15 \text{ K}$ and varies slightly above and below the temperature region. Therefore, the temperature width T_w of the glass transition region was determined as 15 K.

The values of each coefficient at T_g determined by the experiment are shown in Table 1. The values of exponents in each exponential function of T are shown in Table 2. The values of each coefficient at high temperature $T \geq T_g + 15 \text{ K}$ and at low temperature $T \leq T_g - 15 \text{ K}$ are shown in Table 3.

Shape Fixity and Shape Recovery

The stress-strain curves, stress-temperature curves and strain-temperature curves obtained from the thermomechanical test (5) are shown in Figures 5–7, respectively. In

Table 1. Values of coefficients at T_g .

E_g (MPa)	μ_g (GPa·s)	λ_g (s)	C_g	ε_g (%)
146	14	521	0.112	0.3

Table 2. Values of exponents in exponential function of temperature.

a_E	a_μ	a_λ	a_C	a_ε
38.1	44.2	35.4	38.7	58.2

Table 3. Values of coefficients at $T \geq T_g + 15 \text{ K}$ and $T \leq T_g - 15 \text{ K}$.

Coefficients	$T \geq T_g + 15 \text{ K}$	$T \leq T_g - 15 \text{ K}$
E (MPa)	27.6	907
μ (GPa·s)	2.03	116
λ (s)	111	2840
C (—)	0.0206	0.716
ε_l (%)	3.83	0.0184
α (K ⁻¹)	11.6×10^{-5}	

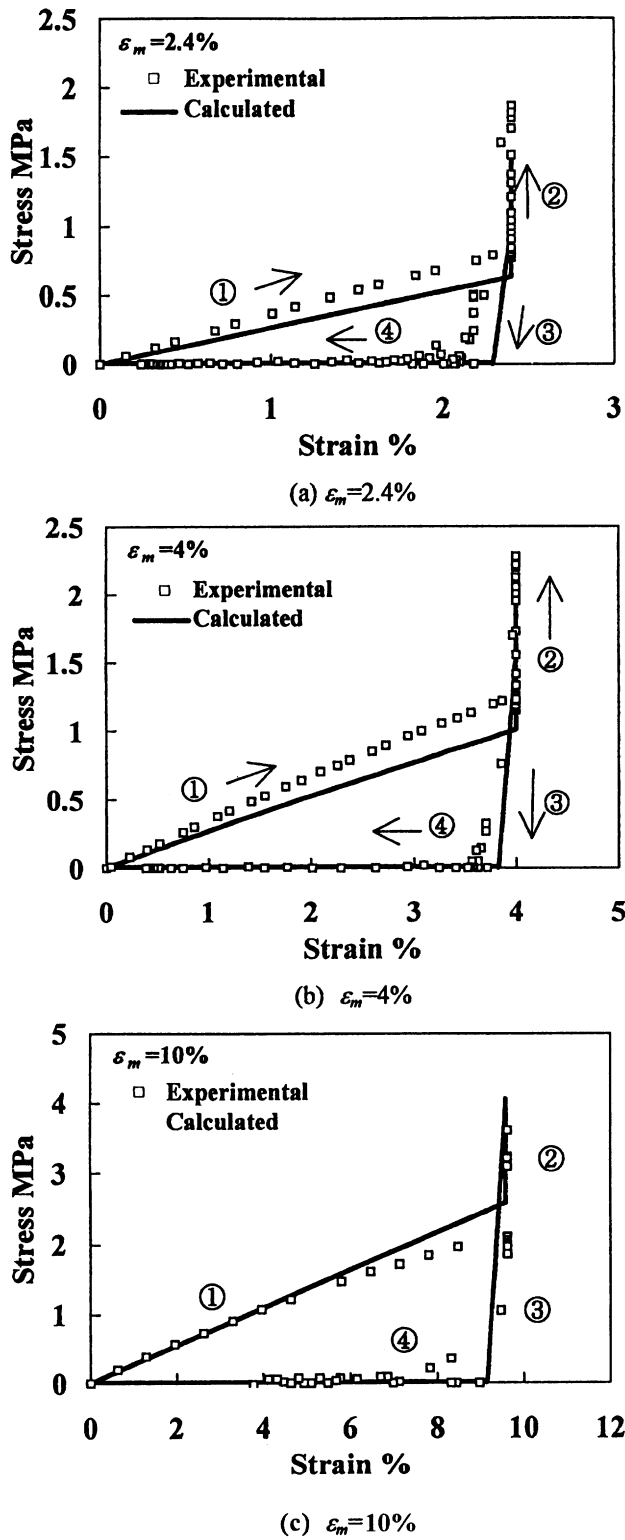


Figure 5. Relationship between stress and strain in the thermomechanical test: (a) $\epsilon_m = 2.4\%$, (b) $\epsilon_m = 4\%$ and (c) $\epsilon_m = 10\%$.

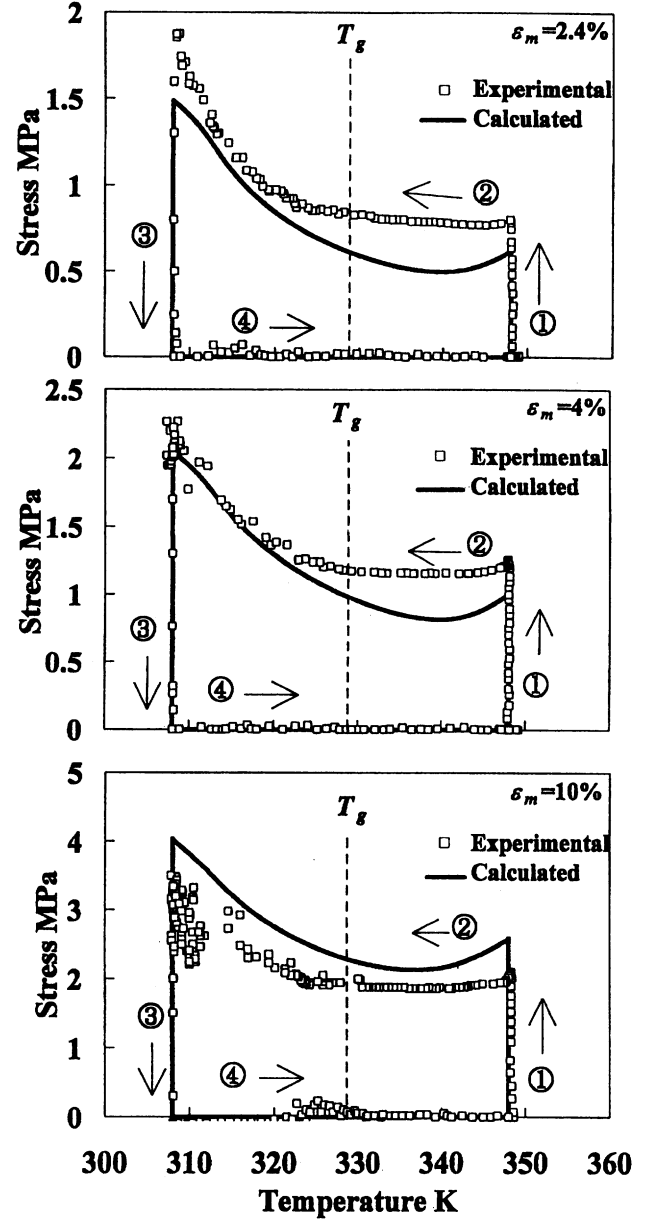


Figure 6. Relationship between stress and temperature in the thermomechanical test.

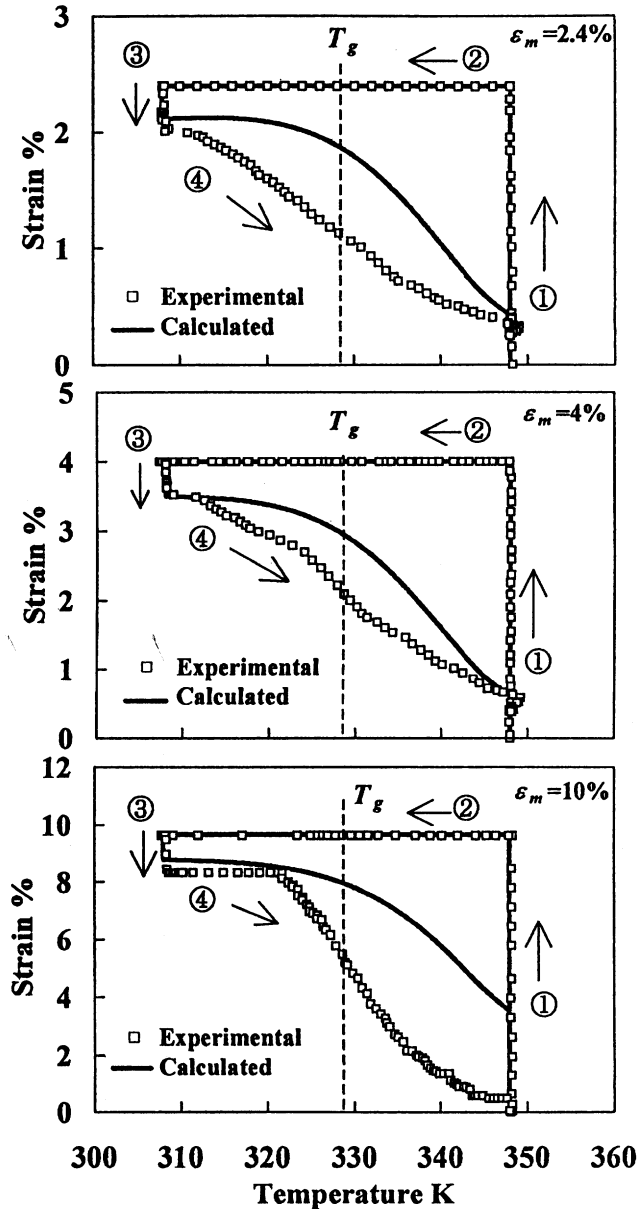


Figure 7. Relationship between strain and temperature in the thermomechanical test.

these figures, symbols ①–④ denote the loading path. In the process ①, the maximum strain ϵ_m was applied at high temperature $T_h = T_g + 20$ K. In the process ②, maintaining ϵ_m constant, the specimen was cooled to $T_l = T_g - 20$ K. In the process ③, the specimen was unloaded at T_l . In the process ④, it was heated from T_l to T_h under the no-load condition. The solid curves denote the calculated results obtained by using the parameters shown in Tables 1–3.

STRESS-STRAIN RELATIONSHIP

As seen in Figure 5, because the maximum strain ϵ_m is smaller than yield strain, the relation is almost proportional in the loading process ①. In the cooling process ②, stress increases. In the unloading process ③, the slopes of the curves are steep and the strain at the unloaded point is close to ϵ_m . This means that the shape fixity of SMP elements is excellent in applications. The reason why the slopes of the curves are different in the processes ① and ③ is due to the fact that elastic modulus is small at temperatures above T_g and large below T_g . In the heating process ④, strain is recovered. The calculated results express the inclination of the experimental results.

STRESS-TEMPERATURE RELATIONSHIP

As seen in Figure 6, during the cooling process ② from T_h to T_l under constant ϵ_m , thermal stress appears against thermal contraction. Stress decreases a little at temperatures in the vicinity of T_h , increases with a decrease in temperature below T_g and takes a maximum value at T_l . The difference $\Delta\sigma = \sigma_l - \sigma_m$ between the maximum stress σ_m in the loading process ① and the maximum stress σ_l in the cooling process ② is as follows. The value of $\Delta\sigma$ is 0.9 MPa for $\epsilon_m = 2.4\%$, 1 MPa for $\epsilon_m = 4\%$ and 1.4 MPa for $\epsilon_m = 10\%$. The dependence of $\Delta\sigma$ on ϵ_m is small. This appears due to the fact that, in these cases, because ϵ_m is small, stress relaxation in the early period of the cooling process is small and the thermal stress $\Delta\sigma_{th} = E\alpha \cdot \Delta T$ due to temperature change ΔT is large. On the contrary, in the case of large ϵ_m beyond yield strain, stress relaxation is large and $\Delta\sigma$ does not appear (Tobushi et al., 1996c). The reason why stress increases significantly at temperatures below T_g in the cooling process ② is due to the fact that E is large at low temperature. The calculated results express the inclination of the experimental results.

STRAIN-TEMPERATURE RELATIONSHIP

As seen in Figure 7, during the heating process ④ from T_l to T_h under no-load condition, strain is recovered. Strain starts to be recovered in the vicinity of $T_g - 15$ K, is significantly recovered in the vicinity of T_g and is recovered gradually above $T_g + 15$ K. The calculated results for strain recovery express the overall inclination of the experimental results. Nevertheless the calculated strain recovery is smaller than the experimental strain recovery. This means the fact that the proposed theory does not precisely express the behavior of strain recovery which accompanies with retardation of time after unloading at T_l .

Recovery Stress

The stress-strain curves and stress-temperature curves obtained by the recovery stress test (6) are shown in Figures 8 and 9, respectively. The symbols ①–⑥ in the figures denote the loading path. The processes ①–③ are the same as those in the thermomechanical test (5). In the process ④, no-load condition was maintained at low temperature T_l for 10 min after unloading. In the process ⑤, the specimen was heated from T_l to high temperature T_h under constant strain. In the process ⑥, it was unloaded at T_h .

STRESS-STRAIN RELATIONSHIP

As seen in Figure 8, the relationship in the processes ①–③ is the same as that in the processes ①–③ shown in Figure 5. In the process ④ under no-load after unloading, strain is recovered by about 1% for 10 min. In the heating process ⑤, stress increases. The slope of the curve in the unloading process ⑥ is almost the same as that in the loading process ①. This appears due to the fact that, because the maximum strain

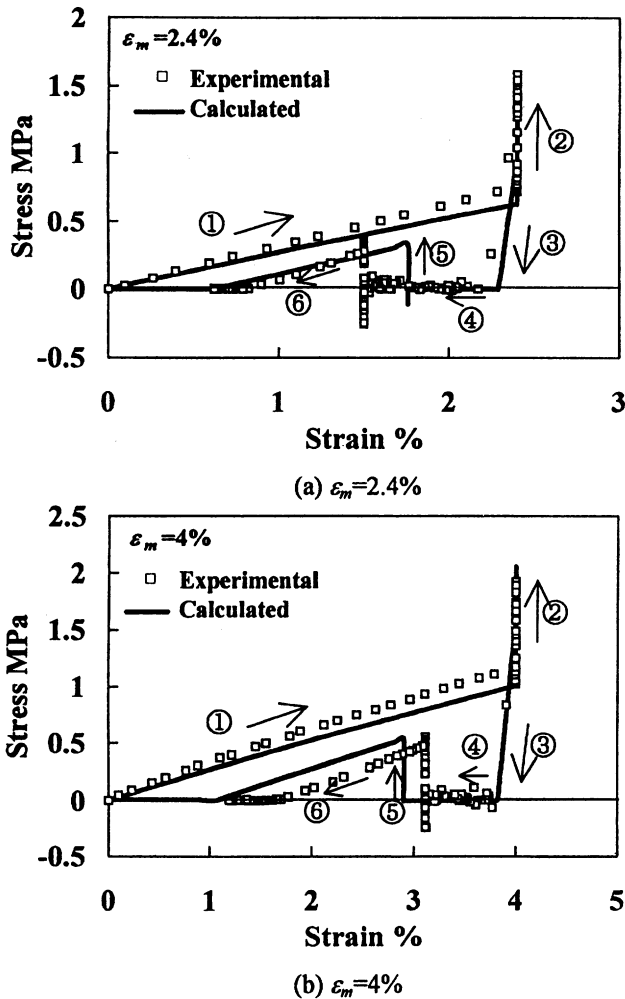


Figure 8. Relationship between stress and strain in the recovery-stress test: (a) $\epsilon_m = 2.4\%$ and (b) $\epsilon_m = 4\%$.

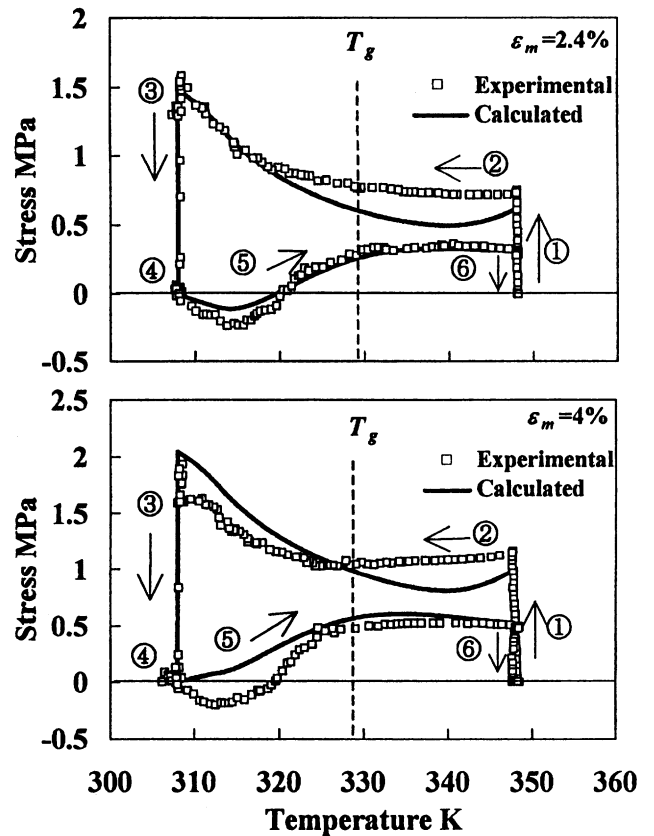


Figure 9. Relationship between stress and temperature in the recovery-stress test.

ϵ_m is below yield strain, deformations under loading and unloading are mainly elastic. In the case of $\epsilon_m = 100\%$, strain is recovered by 1.3% in the process ④. Thus the dependence of strain recovery after unloading at T_l on ϵ_m is small. If ϵ_m is large, the strain ϵ_f obtained by keeping no-load condition after unloading approaches ϵ_m . In many practical applications, we use the shape of SMP elements obtained by deforming at high temperature followed by cooling. Therefore, if ϵ_m is large, ϵ_f obtained at T_l is close to ϵ_m , resulting in excellent shape fixity. The calculated results express the inclination of the experimental results.

STRESS-TEMPERATURE RELATIONSHIP

As seen in Figure 9, the relationship in the processes ①–③ is the same as that in the processes ①–③ shown in Figure 6. In the heating process under constant strain ⑤ which follows the process ④, stress decreases in the vicinity of T_l , increases around T_g and is almost constant above T_g . If there is not the process ④, that is, if the specimen is heated just after unloading without having the time under no-load at T_h , stress increases immediately and does not decrease by heating. This behavior of stress without the process ④ will be discussed in the following report. The decrease in stress which is observed in the early stage of heating process ⑤ appears due to compressive stress against thermal expansion. The increase in stress around T_g appears against the constraint of strain be-

cause the specimen has the inclination to be recovered due to the micro-Brownian motion of molecular chain activated by heating. If there is not the constraint of strain, strain is recovered without increase in stress by heating as observed in Figure 7. Although recovery stress increases around T_g and is almost constant thereafter, recovery stress decreases a little in the vicinity of T_h . The decrease in stress appears due to stress relaxation at high temperature. The calculated results express the overall inclination of the experimental results. In the case of $\varepsilon_m = 4\%$, the decrease in stress in the vicinity of T_l during the heating process ⑤ cannot be expressed precisely. This discrepancy appears due to the fact that the estimated recovery stress is larger than the compressive thermal stress in the case of large ε_m .

CONCLUSIONS

In order to describe the thermomechanical properties in shape memory polymer of polyurethane series, a thermomechanical constitutive model was developed by modifying a standard linear viscoelastic model. The model involved a slip element due to internal friction and took account of thermal expansion. In order to describe the variation in mechanical properties due to the glass transition, coefficients in the model were expressed by a single exponential function of temperature. Several kinds of thermomechanical tests were carried out. The proposed theory expressed well the thermomechanical properties of the material, such as shape fix-

ity, shape recovery and recovery stress. The proposed model is useful for design of shape-memory polymer elements, in which the amount of recovery deformation, the tightening force and the working start and completion temperatures are specified.

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